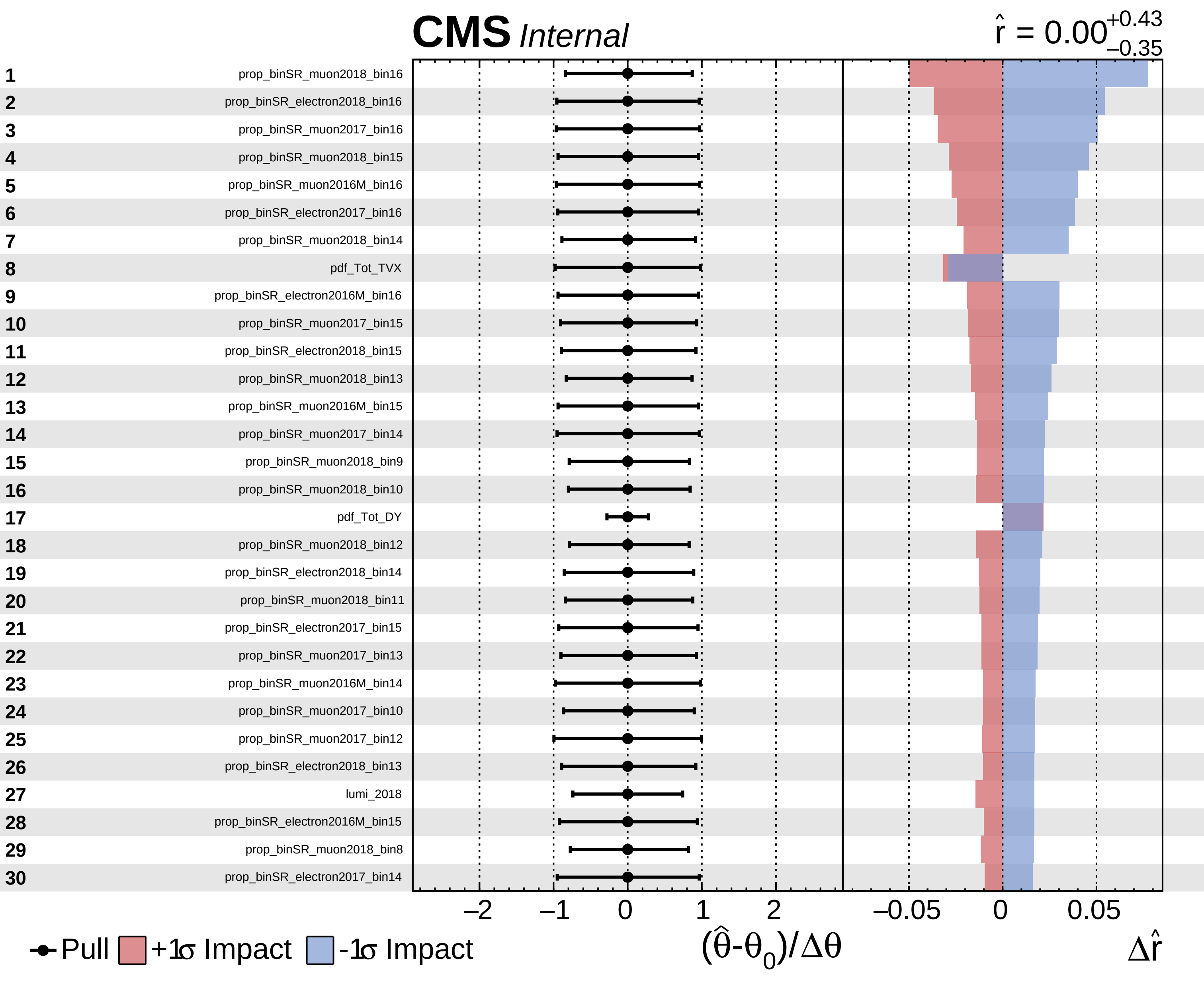
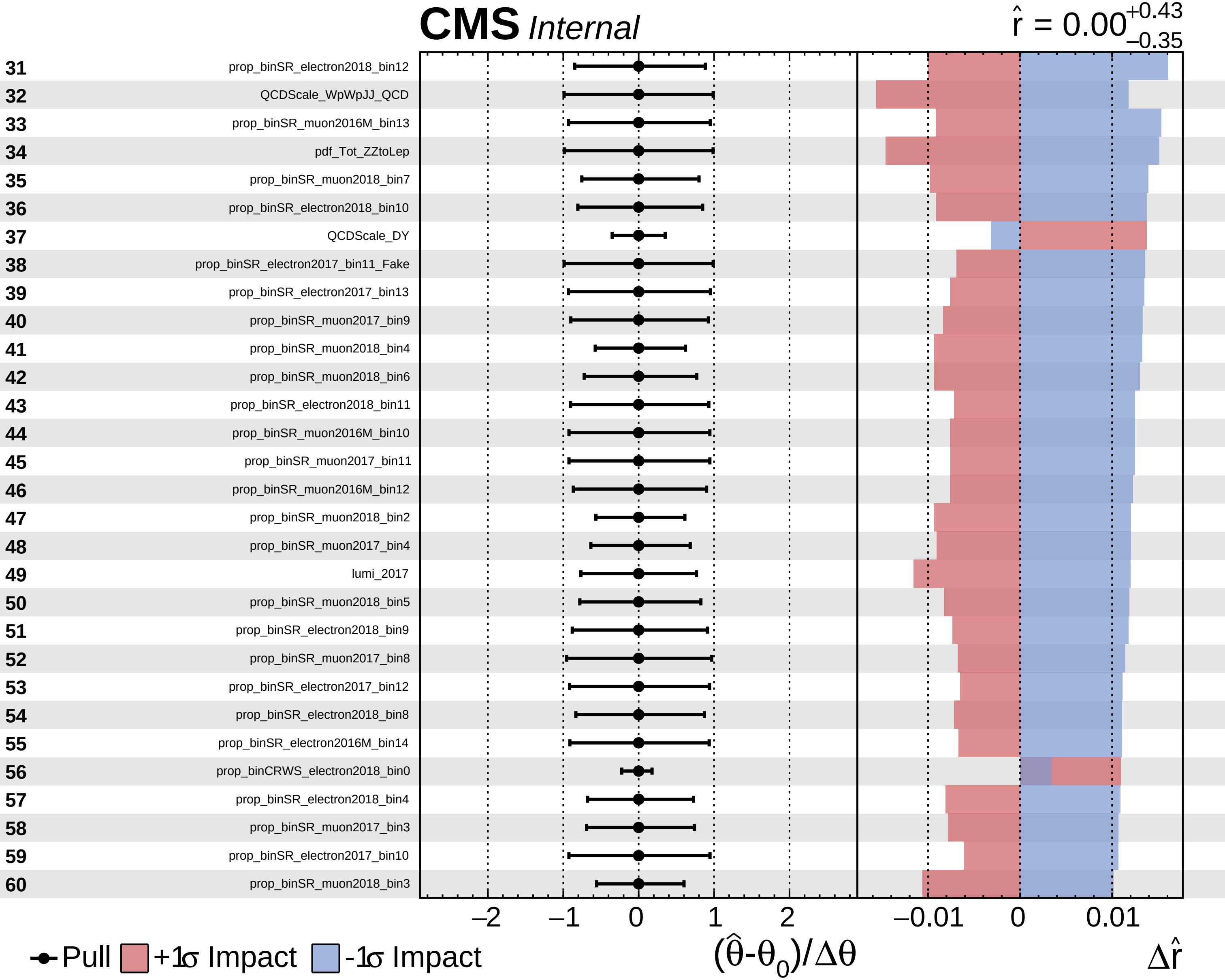
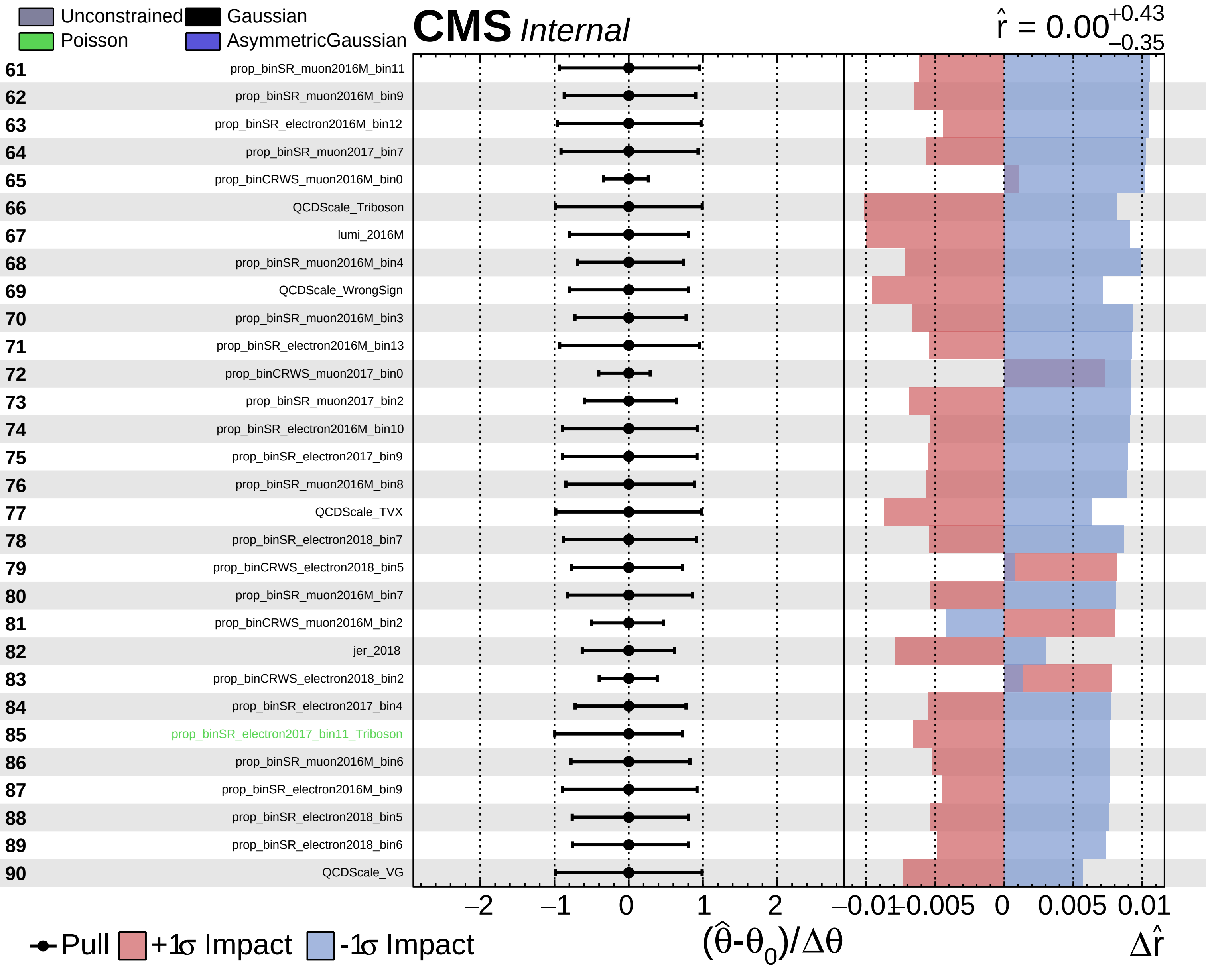
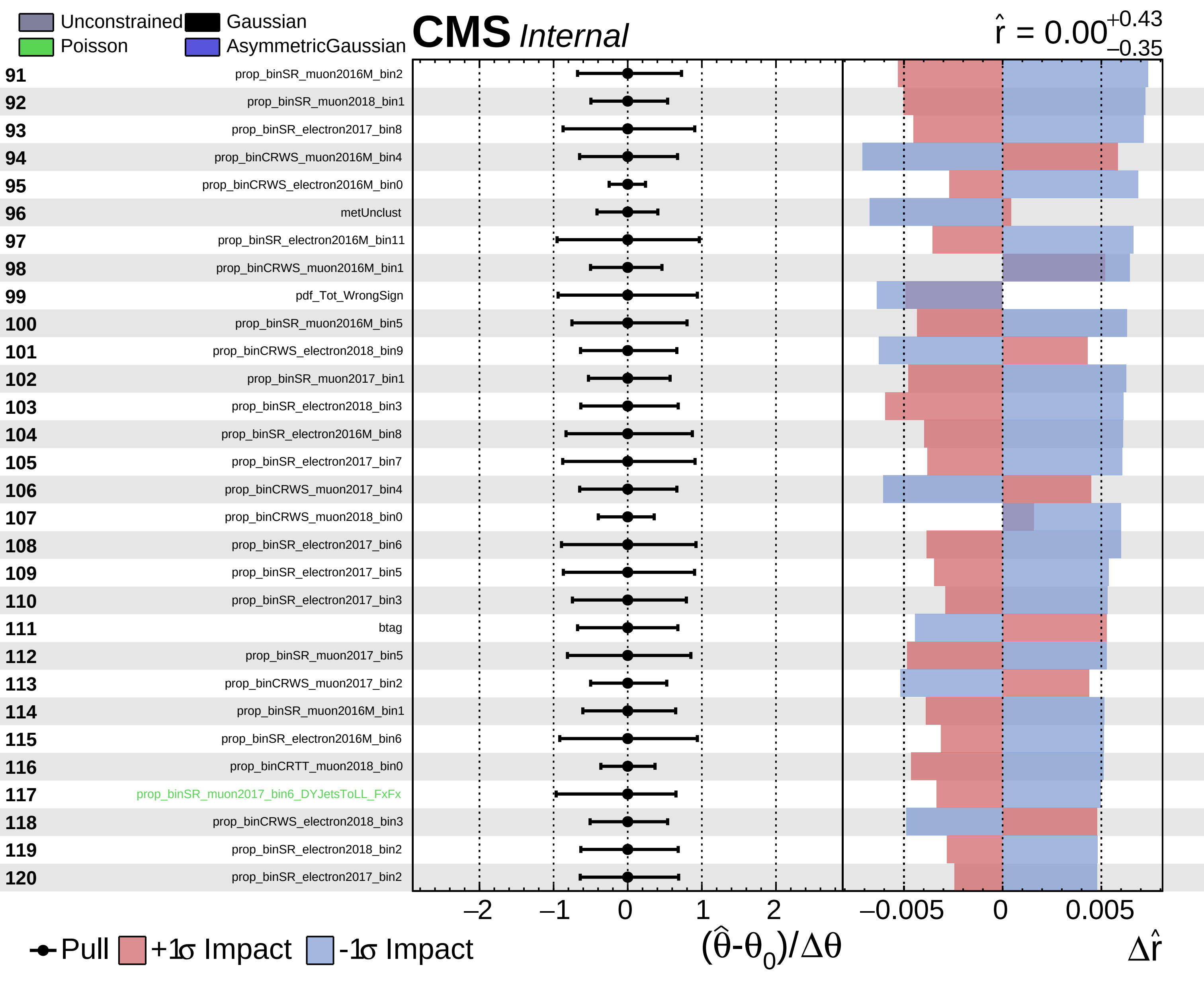
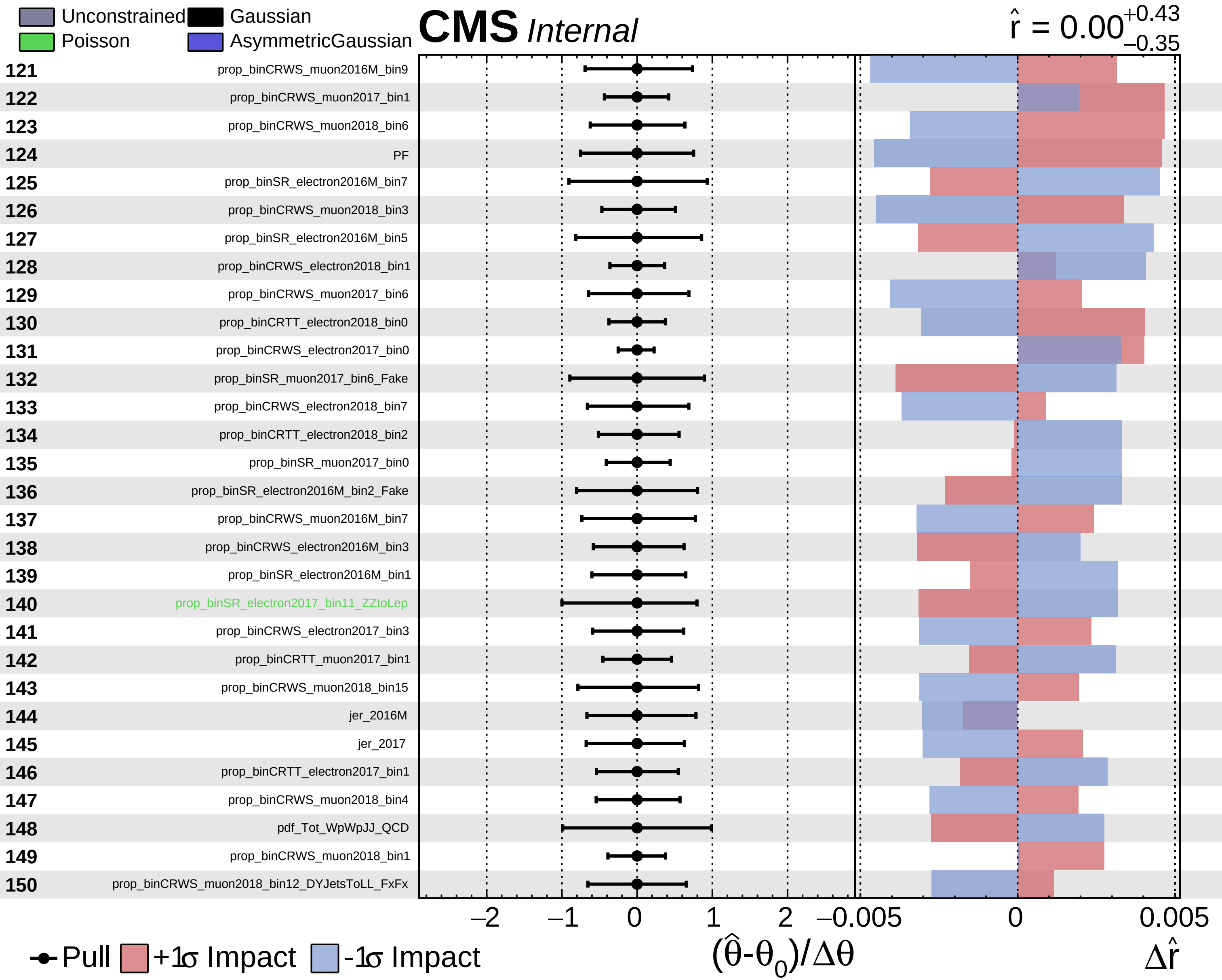

CMS Internal
$\hat{r} = 0.00^{+0.43}_{-0.35}$
1 prop_binSR_muon2018_bin16
2 prop_binSR_electron2018_bin16
3 prop_binSR_muon2017_bin16
4 prop_binSR_muon2018_bin15
5 prop_binSR_muon2016M_bin16
6 prop_binSR_electron2017_bin16
7 prop_binSR_muon2018_bin14
8 pdf_Tot_TVX
9 prop_binSR_electron2016M_bin16
10 prop_binSR_muon2017_bin15
11 prop_binSR_electron2018_bin15
12 prop_binSR_muon2018_bin13
13 prop_binSR_muon2016M_bin15
14 prop_binSR_muon2017_bin14
15 prop_binSR_muon2018_bin9
16 prop_binSR_muon2018_bin10
17 pdf_Tot_DY
18 prop_binSR_muon2018_bin12
19 prop_binSR_electron2018_bin14
20 prop_binSR_muon2018_bin11
21 prop_binSR_electron2017_bin15
22 prop_binSR_muon2017_bin13
23 prop_binSR_muon2016M_bin14
24 prop_binSR_muon2017_bin10
25 prop_binSR_muon2017_bin12
26 prop_binSR_electron2018_bin13
27 lumi_2018
28 prop_binSR_electron2016M_bin15
29 prop_binSR_muon2018_bin8
30 prop_binSR_electron2017_bin14
−2 −1 0 1 2
−0.05 0 0.05
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



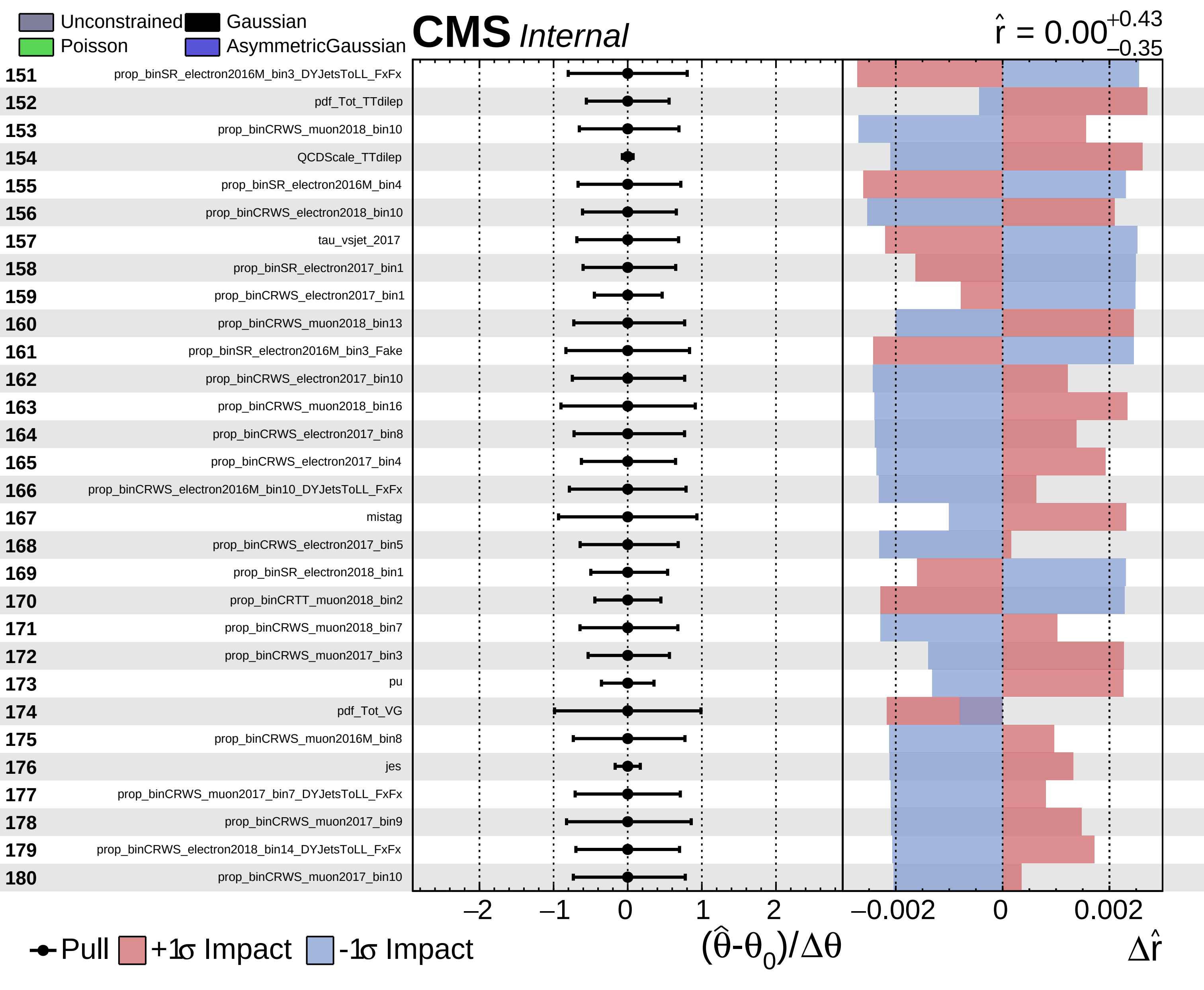
CMS Internal
$\hat{r} = 0.00^{+0.43}_{-0.35}$
31 prop_binSR_electron2018_bin12
32 QCDScale_WpWpJJ_QCD
33 prop_binSR_muon2016M_bin13
34 pdf_Tot_ZZtoLep
35 prop_binSR_muon2018_bin7
36 prop_binSR_electron2018_bin10
37 QCDScale_DY
38 prop_binSR_electron2017_bin11_Fake
39 prop_binSR_electron2017_bin13
40 prop_binSR_muon2017_bin9
41 prop_binSR_muon2018_bin4
42 prop_binSR_muon2018_bin6
43 prop_binSR_electron2018_bin11
44 prop_binSR_muon2016M_bin10
45 prop_binSR_muon2017_bin11
46 prop_binSR_muon2016M_bin12
47 prop_binSR_muon2018_bin2
48 prop_binSR_muon2017_bin4
49 lumi_2017
50 prop_binSR_muon2018_bin5
51 prop_binSR_electron2018_bin9
52 prop_binSR_muon2017_bin8
53 prop_binSR_electron2017_bin12
54 prop_binSR_electron2018_bin8
55 prop_binSR_electron2016M_bin14
56 prop_binCRWS_electron2018_bin0
57 prop_binSR_electron2018_bin4
58 prop_binSR_muon2017_bin3
59 prop_binSR_electron2017_bin10
60 prop_binSR_muon2018_bin3
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

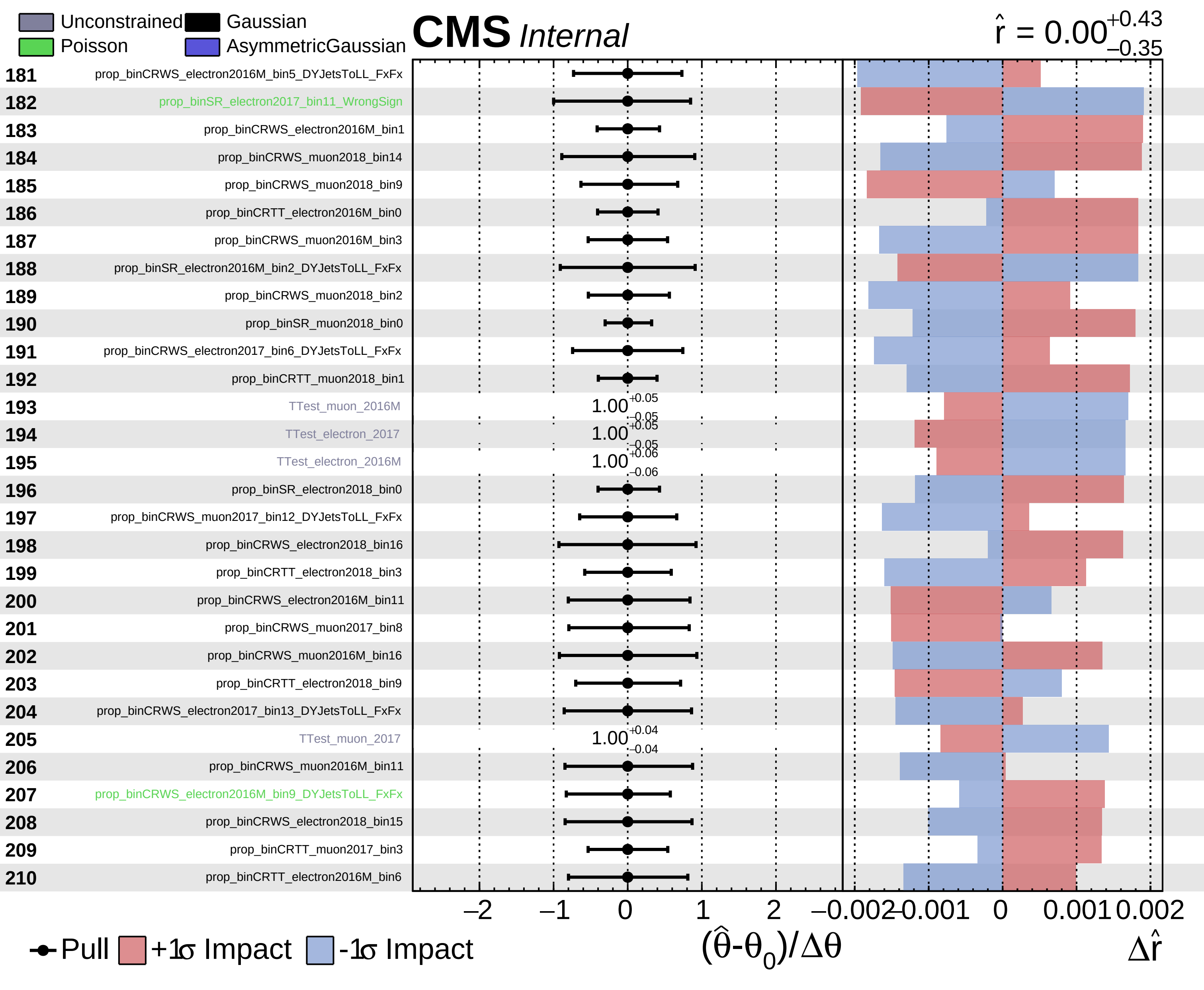

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.43}_{-0.35}$
61 prop_binSR_muon2016M_bin11
62 prop_binSR_muon2016M_bin9
63 prop_binSR_electron2016M_bin12
64 prop_binSR_muon2017_bin7
65 prop_binCRWS_muon2016M_bin0
66 QCDScale_Triboson
67 lumi_2016M
68 prop_binSR_muon2016M_bin4
69 QCDScale_WrongSign
70 prop_binSR_muon2016M_bin3
71 prop_binSR_electron2016M_bin13
72 prop_binCRWS_muon2017_bin0
73 prop_binSR_muon2017_bin2
74 prop_binSR_electron2016M_bin10
75 prop_binSR_electron2017_bin9
76 prop_binSR_muon2016M_bin8
77 QCDScale_TVX
78 prop_binSR_electron2018_bin7
79 prop_binCRWS_electron2018_bin5
80 prop_binSR_muon2016M_bin7
81 prop_binCRWS_muon2016M_bin2
82 jer_2018
83 prop_binCRWS_electron2018_bin2
84 prop_binSR_electron2017_bin4
85 prop_binSR_electron2017_bin11_Triboson
86 prop_binSR_muon2016M_bin6
87 prop_binSR_electron2016M_bin9
88 prop_binSR_electron2018_bin5
89 prop_binSR_electron2018_bin6
90 QCDScale_VG
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.43 −0.35
91 prop_binSR_muon2016M_bin2
92 prop_binSR_muon2018_bin1
93 prop_binSR_electron2017_bin8
94 prop_binCRWS_muon2016M_bin4
95 prop_binCRWS_electron2016M_bin0
96 metUnclust
97 prop_binSR_electron2016M_bin11
98 prop_binCRWS_muon2016M_bin1
99 pdf_Tot_WrongSign
100 prop_binSR_muon2016M_bin5
101 prop_binCRWS_electron2018_bin9
102 prop_binSR_muon2017_bin1
103 prop_binSR_electron2018_bin3
104 prop_binSR_electron2016M_bin8
105 prop_binSR_electron2017_bin7
106 prop_binCRWS_muon2017_bin4
107 prop_binCRWS_muon2018_bin0
108 prop_binSR_electron2017_bin6
109 prop_binSR_electron2017_bin5
110 prop_binSR_electron2017_bin3
111 btag
112 prop_binSR_muon2017_bin5
113 prop_binCRWS_muon2017_bin2
114 prop_binSR_muon2016M_bin1
115 prop_binSR_electron2016M_bin6
116 prop_binCRTT_muon2018_bin0
117 prop_binSR_muon2017_bin6_DYJetsToLL_FxFx
118 prop_binCRWS_electron2018_bin3
119 prop_binSR_electron2018_bin2
120 prop_binSR_electron2017_bin2
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

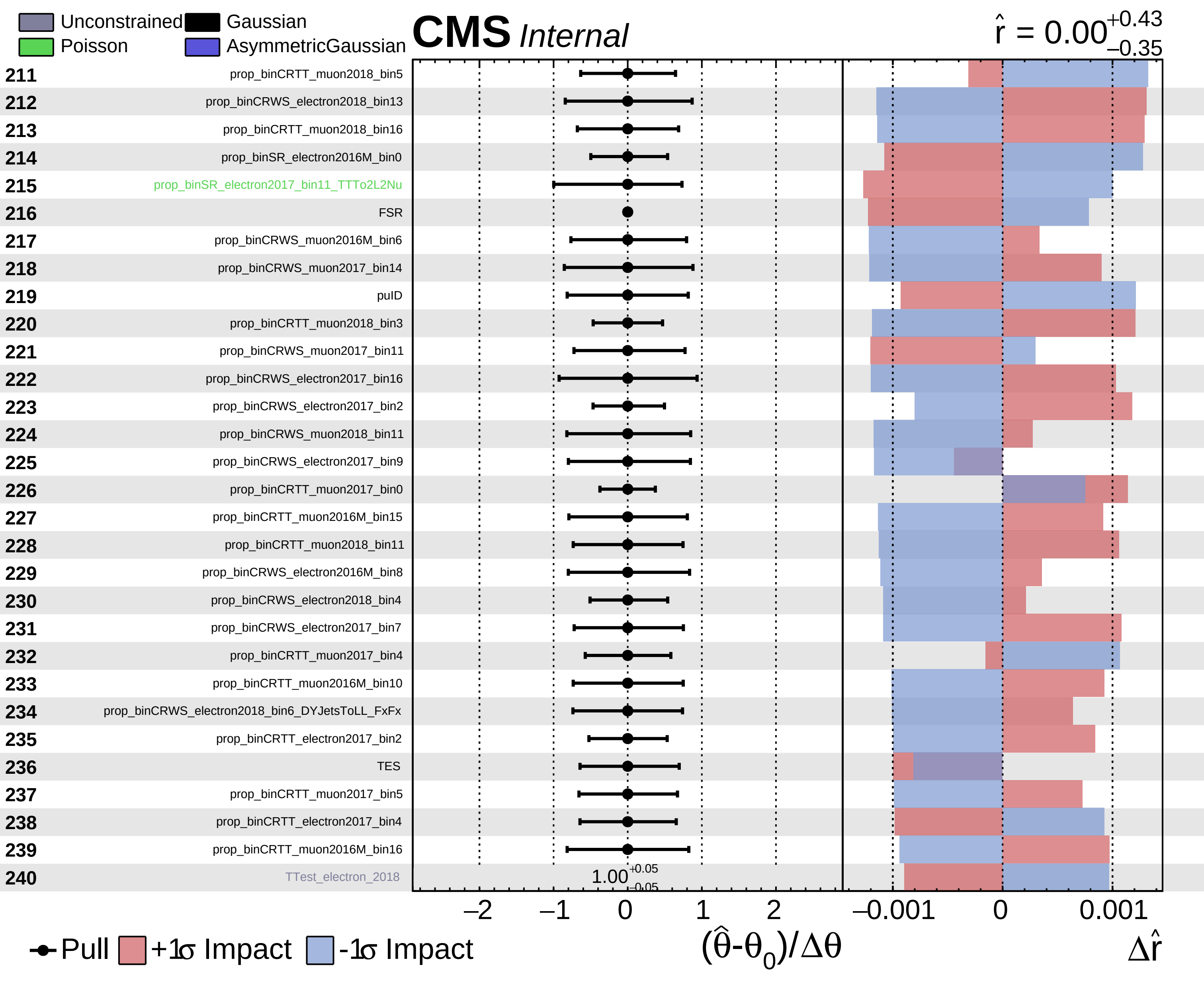


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.43}_{-0.35}$
121 prop_binCRWS_muon2016M_bin9
122 prop_binCRWS_muon2017_bin1
123 prop_binCRWS_muon2018_bin6
124 PF
125 prop_binSR_electron2016M_bin7
126 prop_binCRWS_muon2018_bin3
127 prop_binSR_electron2016M_bin5
128 prop_binCRWS_electron2018_bin1
129 prop_binCRWS_muon2017_bin6
130 prop_binCRTT_electron2018_bin0
131 prop_binCRWS_electron2017_bin0
132 prop_binSR_muon2017_bin6_Fake
133 prop_binCRWS_electron2018_bin7
134 prop_binCRTT_electron2018_bin2
135 prop_binSR_muon2017_bin0
136 prop_binSR_electron2016M_bin2_Fake
137 prop_binCRWS_muon2016M_bin7
138 prop_binCRWS_electron2016M_bin3
139 prop_binSR_electron2016M_bin1
140 prop_binSR_electron2017_bin11_ZZtoLep
141 prop_binCRWS_electron2017_bin3
142 prop_binCRTT_muon2017_bin1
143 prop_binCRWS_muon2018_bin15
144 jer_2016M
145 jer_2017
146 prop_binCRTT_electron2017_bin1
147 prop_binCRWS_muon2018_bin4
148 pdf_Tot_WpWpJJ_QCD
149 prop_binCRWS_muon2018_bin1
150 prop_binCRWS_muon2018_bin12_DYJetsToLL_FxFx
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

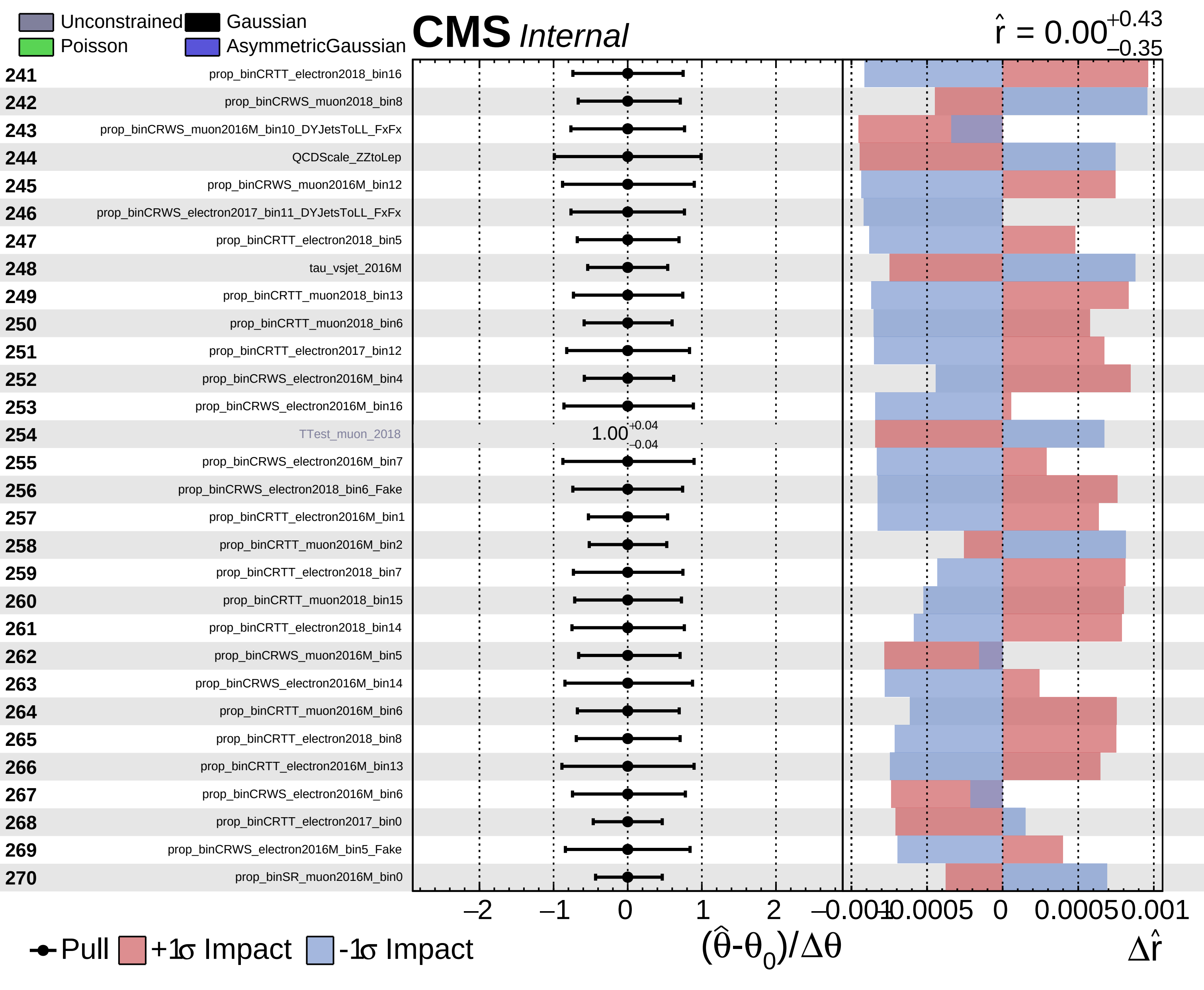

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.43 −0.35
151 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
152 pdf_Tot_TTdilep
153 prop_binCRWS_muon2018_bin10
154 QCDScale_TTdilep
155 prop_binSR_electron2016M_bin4
156 prop_binCRWS_electron2018_bin10
157 tau_vsjet_2017
158 prop_binSR_electron2017_bin1
159 prop_binCRWS_electron2017_bin1
160 prop_binCRWS_muon2018_bin13
161 prop_binSR_electron2016M_bin3_Fake
162 prop_binCRWS_electron2017_bin10
163 prop_binCRWS_muon2018_bin16
164 prop_binCRWS_electron2017_bin8
165 prop_binCRWS_electron2017_bin4
166 prop_binCRWS_electron2016M_bin10_DYJetsToLL_FxFx
167 mistag
168 prop_binCRWS_electron2017_bin5
169 prop_binSR_electron2018_bin1
170 prop_binCRTT_muon2018_bin2
171 prop_binCRWS_muon2018_bin7
172 prop_binCRWS_muon2017_bin3
173 pu
174 pdf_Tot_VG
175 prop_binCRWS_muon2016M_bin8
176 jes
177 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
178 prop_binCRWS_muon2017_bin9
179 prop_binCRWS_electron2018_bin14_DYJetsToLL_FxFx
180 prop_binCRWS_muon2017_bin10
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



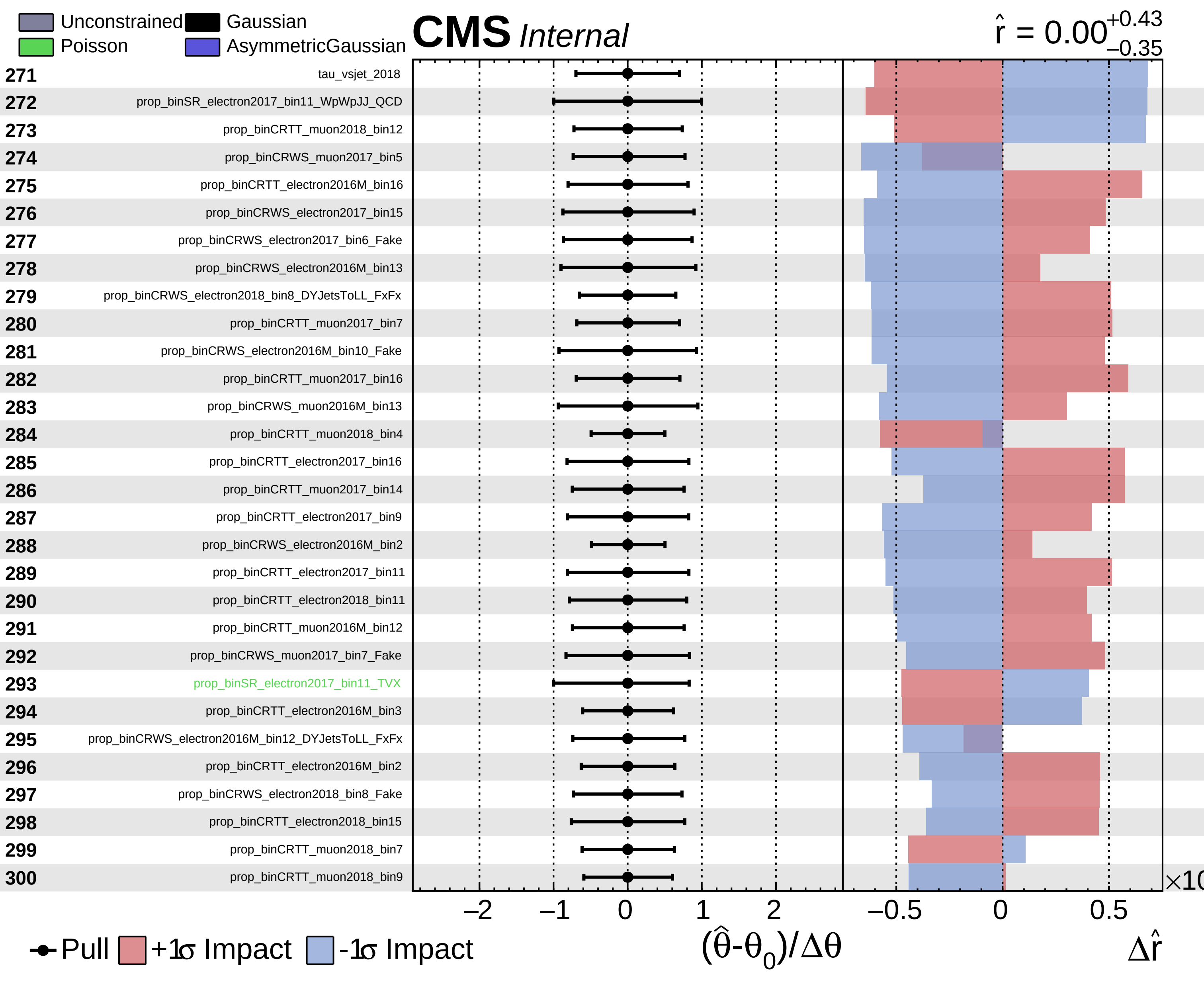
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.43}_{-0.35}$
181 prop_binCRWS_electron2016M_bin5_DYJetsToLL_FxFx
182 prop_binSR_electron2017_bin11_WrongSign
183 prop_binCRWS_electron2016M_bin1
184 prop_binCRWS_muon2018_bin14
185 prop_binCRWS_muon2018_bin9
186 prop_binCRTT_electron2016M_bin0
187 prop_binCRWS_muon2016M_bin3
188 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
189 prop_binCRWS_muon2018_bin2
190 prop_binSR_muon2018_bin0
191 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
192 prop_binCRTT_muon2018_bin1
193 TTest_muon_2016M $1.00^{+0.05}_{-0.05}$
194 TTest_electron_2017 $1.00^{+0.05}_{-0.05}$
195 TTest_electron_2016M $1.00^{+0.06}_{-0.06}$
196 prop_binSR_electron2018_bin0
197 prop_binCRWS_muon2017_bin12_DYJetsToLL_FxFx
198 prop_binCRWS_electron2018_bin16
199 prop_binCRTT_electron2018_bin3
200 prop_binCRWS_electron2016M_bin11
201 prop_binCRWS_muon2017_bin8
202 prop_binCRWS_muon2016M_bin16
203 prop_binCRTT_electron2018_bin9
204 prop_binCRWS_electron2017_bin13_DYJetsToLL_FxFx
205 TTest_muon_2017 $1.00^{+0.04}_{-0.04}$
206 prop_binCRWS_muon2016M_bin11
207 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
208 prop_binCRWS_electron2018_bin15
209 prop_binCRTT_muon2017_bin3
210 prop_binCRTT_electron2016M_bin6
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

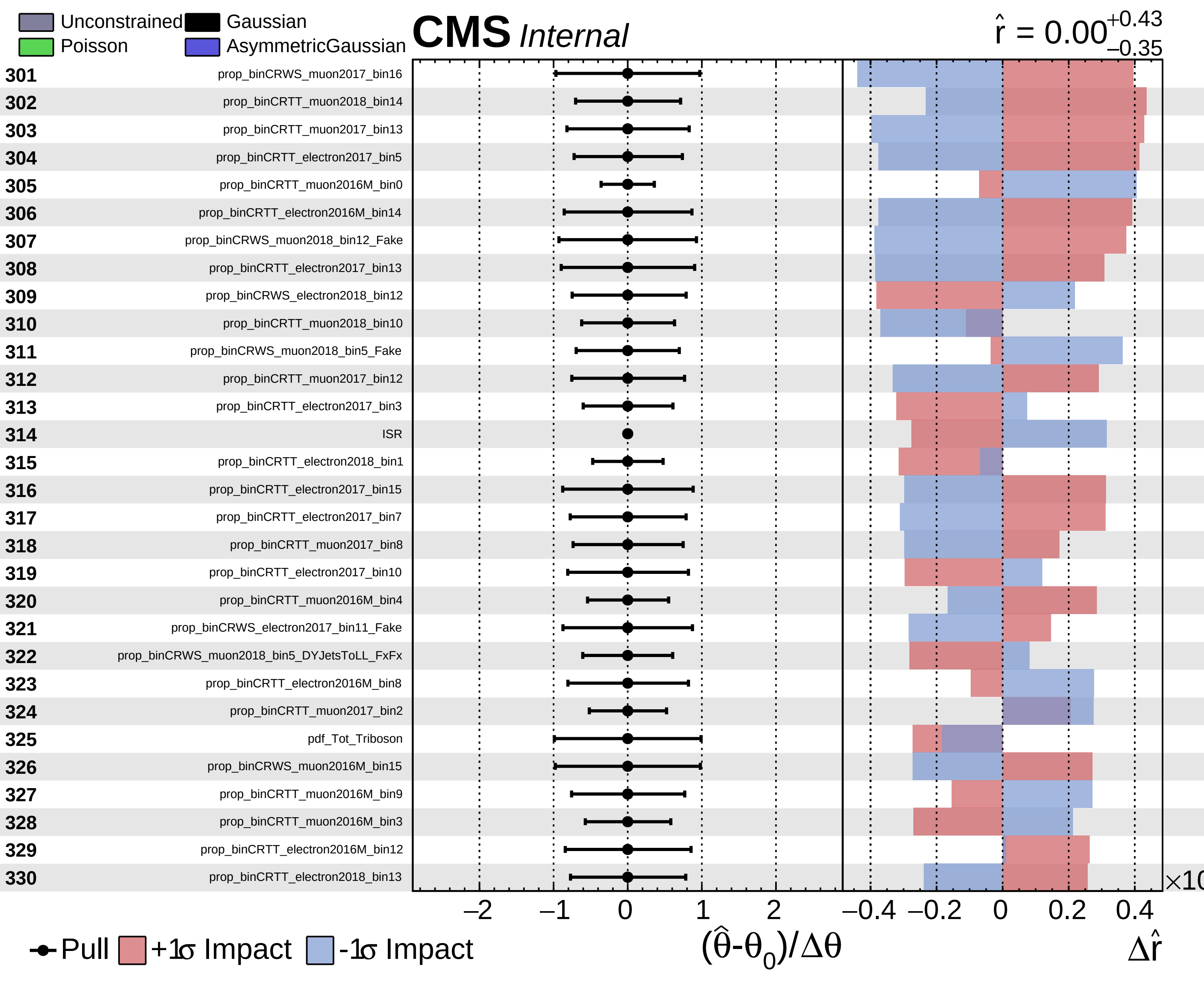


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.43}_{-0.35}$
211 prop_binCRTT_muon2018_bin5
212 prop_binCRWS_electron2018_bin13
213 prop_binCRTT_muon2018_bin16
214 prop_binSR_electron2016M_bin0
215 prop_binSR_electron2017_bin11_TTTo2L2Nu
216 FSR
217 prop_binCRWS_muon2016M_bin6
218 prop_binCRWS_muon2017_bin14
219 puID
220 prop_binCRTT_muon2018_bin3
221 prop_binCRWS_muon2017_bin11
222 prop_binCRWS_electron2017_bin16
223 prop_binCRWS_electron2017_bin2
224 prop_binCRWS_muon2018_bin11
225 prop_binCRWS_electron2017_bin9
226 prop_binCRTT_muon2017_bin0
227 prop_binCRTT_muon2016M_bin15
228 prop_binCRTT_muon2018_bin11
229 prop_binCRWS_electron2016M_bin8
230 prop_binCRWS_electron2018_bin4
231 prop_binCRWS_electron2017_bin7
232 prop_binCRTT_muon2017_bin4
233 prop_binCRTT_muon2016M_bin10
234 prop_binCRWS_electron2018_bin6_DYJetsToLL_FxFx
235 prop_binCRTT_electron2017_bin2
236 TES
237 prop_binCRTT_muon2017_bin5
238 prop_binCRTT_electron2017_bin4
239 prop_binCRTT_muon2016M_bin16
240 TTest_electron_2018
$1.00^{+0.05}_{-0.05}$
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

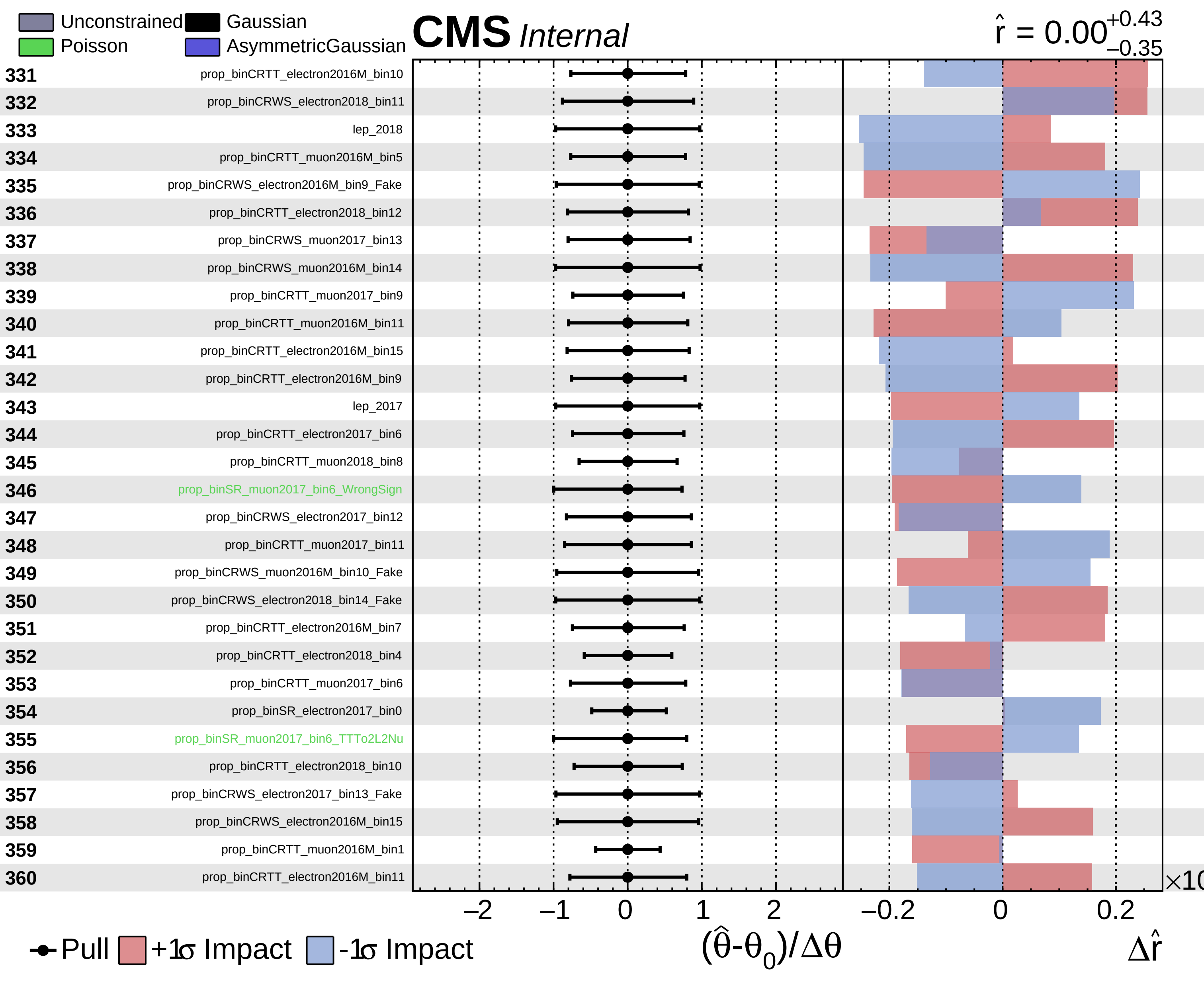


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.43}_{-0.35}$
241 prop_binCRTT_electron2018_bin16
242 prop_binCRWS_muon2018_bin8
243 prop_binCRWS_muon2016M_bin10_DYJetsToLL_FxFx
244 QCDScale_ZZtoLep
245 prop_binCRWS_muon2016M_bin12
246 prop_binCRWS_electron2017_bin11_DYJetsToLL_FxFx
247 prop_binCRTT_electron2018_bin5
248 tau_vsjet_2016M
249 prop_binCRTT_muon2018_bin13
250 prop_binCRTT_muon2018_bin6
251 prop_binCRTT_electron2017_bin12
252 prop_binCRWS_electron2016M_bin4
253 prop_binCRWS_electron2016M_bin16
254 TTest_muon_2018
$1.00^{+0.04}_{-0.04}$
255 prop_binCRWS_electron2016M_bin7
256 prop_binCRWS_electron2018_bin6_Fake
257 prop_binCRTT_electron2016M_bin1
258 prop_binCRTT_muon2016M_bin2
259 prop_binCRTT_electron2018_bin7
260 prop_binCRTT_muon2018_bin15
261 prop_binCRTT_electron2018_bin14
262 prop_binCRWS_muon2016M_bin5
263 prop_binCRWS_electron2016M_bin14
264 prop_binCRTT_muon2016M_bin6
265 prop_binCRTT_electron2018_bin8
266 prop_binCRTT_electron2016M_bin13
267 prop_binCRWS_electron2016M_bin6
268 prop_binCRTT_electron2017_bin0
269 prop_binCRWS_electron2016M_bin5_Fake
270 prop_binSR_muon2016M_bin0
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

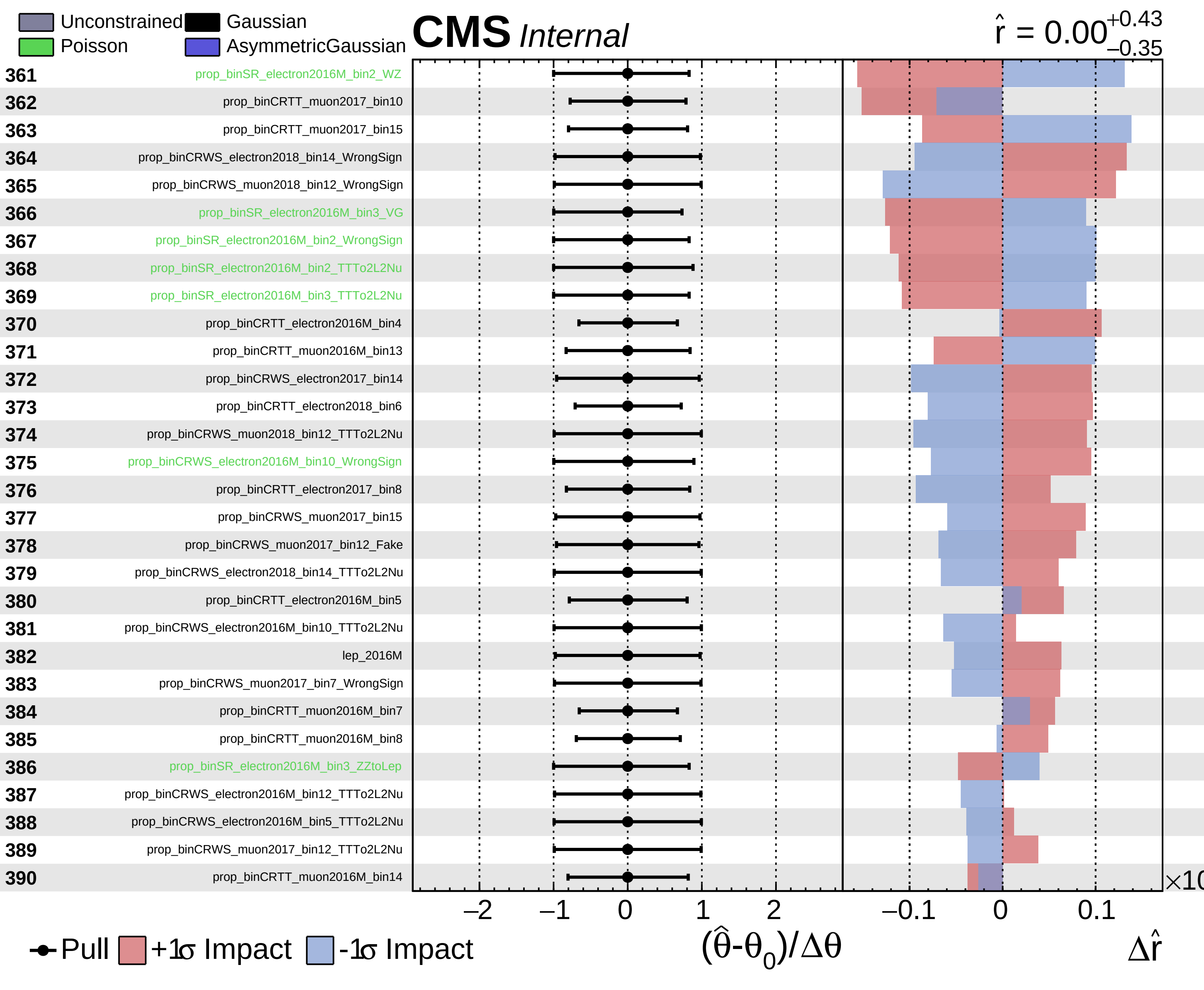

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.43}_{-0.35}$
271 tau_vsjet_2018
272 prop_binSR_electron2017_bin11_WpWpJJ_QCD
273 prop_binCRTT_muon2018_bin12
274 prop_binCRWS_muon2017_bin5
275 prop_binCRTT_electron2016M_bin16
276 prop_binCRWS_electron2017_bin15
277 prop_binCRWS_electron2017_bin6_Fake
278 prop_binCRWS_electron2016M_bin13
279 prop_binCRWS_electron2018_bin8_DYJetsToLL_FxFx
280 prop_binCRTT_muon2017_bin7
281 prop_binCRWS_electron2016M_bin10_Fake
282 prop_binCRTT_muon2017_bin16
283 prop_binCRWS_muon2016M_bin13
284 prop_binCRTT_muon2018_bin4
285 prop_binCRTT_electron2017_bin16
286 prop_binCRTT_muon2017_bin14
287 prop_binCRTT_electron2017_bin9
288 prop_binCRWS_electron2016M_bin2
289 prop_binCRTT_electron2017_bin11
290 prop_binCRTT_electron2018_bin11
291 prop_binCRTT_muon2016M_bin12
292 prop_binCRWS_muon2017_bin7_Fake
293 prop_binSR_electron2017_bin11_TVX
294 prop_binCRTT_electron2016M_bin3
295 prop_binCRWS_electron2016M_bin12_DYJetsToLL_FxFx
296 prop_binCRTT_electron2016M_bin2
297 prop_binCRWS_electron2018_bin8_Fake
298 prop_binCRTT_electron2018_bin15
299 prop_binCRTT_muon2018_bin7
300 prop_binCRTT_muon2018_bin9
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



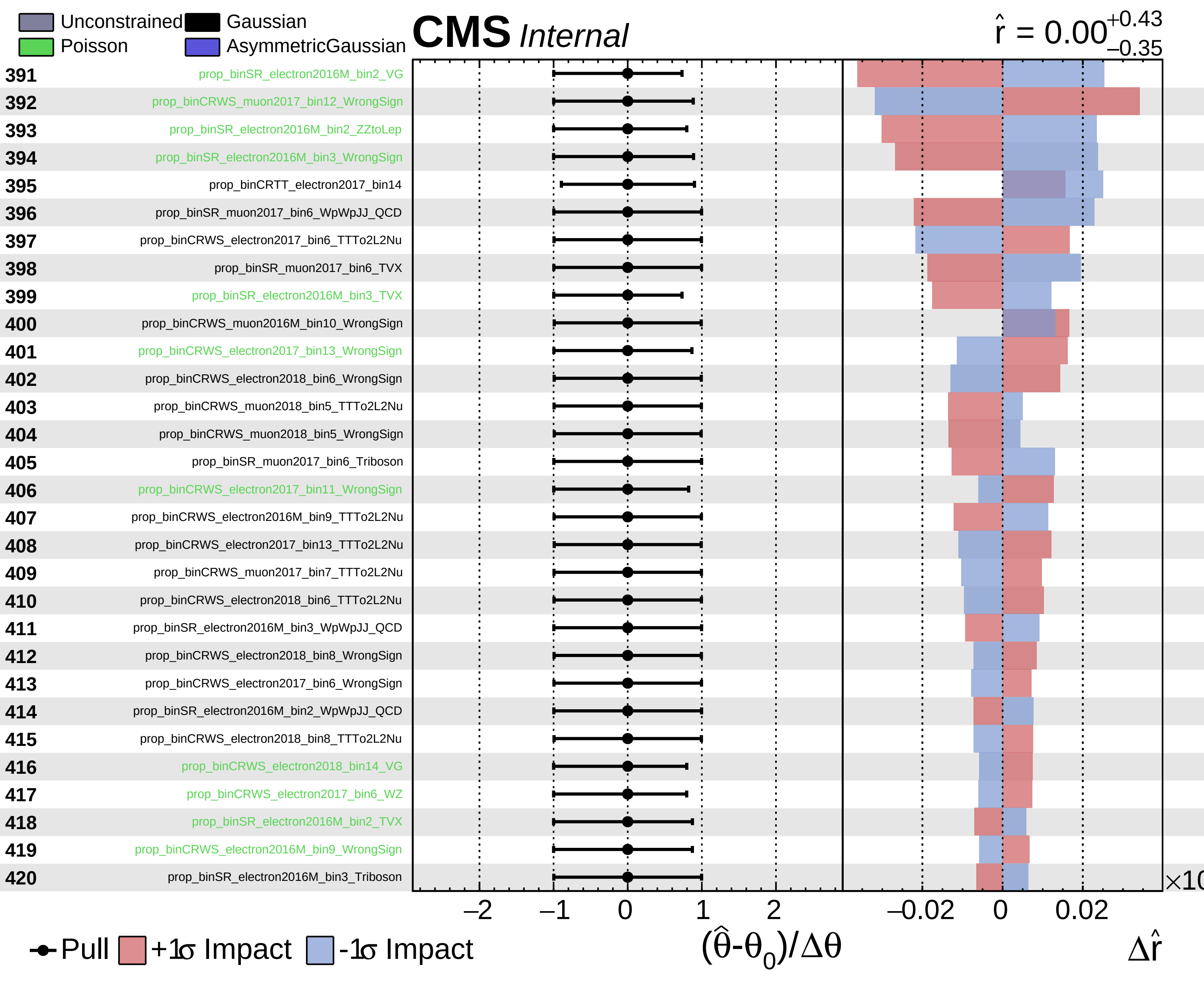
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.43}_{-0.35}$
301 prop_binCRWS_muon2017_bin16
302 prop_binCRTT_muon2018_bin14
303 prop_binCRTT_muon2017_bin13
304 prop_binCRTT_electron2017_bin5
305 prop_binCRTT_muon2016M_bin0
306 prop_binCRTT_electron2016M_bin14
307 prop_binCRWS_muon2018_bin12_Fake
308 prop_binCRTT_electron2017_bin13
309 prop_binCRWS_electron2018_bin12
310 prop_binCRTT_muon2018_bin10
311 prop_binCRWS_muon2018_bin5_Fake
312 prop_binCRTT_muon2017_bin12
313 prop_binCRTT_electron2017_bin3
314 ISR
315 prop_binCRTT_electron2018_bin1
316 prop_binCRTT_electron2017_bin15
317 prop_binCRTT_electron2017_bin7
318 prop_binCRTT_muon2017_bin8
319 prop_binCRTT_electron2017_bin10
320 prop_binCRTT_muon2016M_bin4
321 prop_binCRWS_electron2017_bin11_Fake
322 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
323 prop_binCRTT_electron2016M_bin8
324 prop_binCRTT_muon2017_bin2
325 pdf_Tot_Triboson
326 prop_binCRWS_muon2016M_bin15
327 prop_binCRTT_muon2016M_bin9
328 prop_binCRTT_muon2016M_bin3
329 prop_binCRTT_electron2016M_bin12
330 prop_binCRTT_electron2018_bin13
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



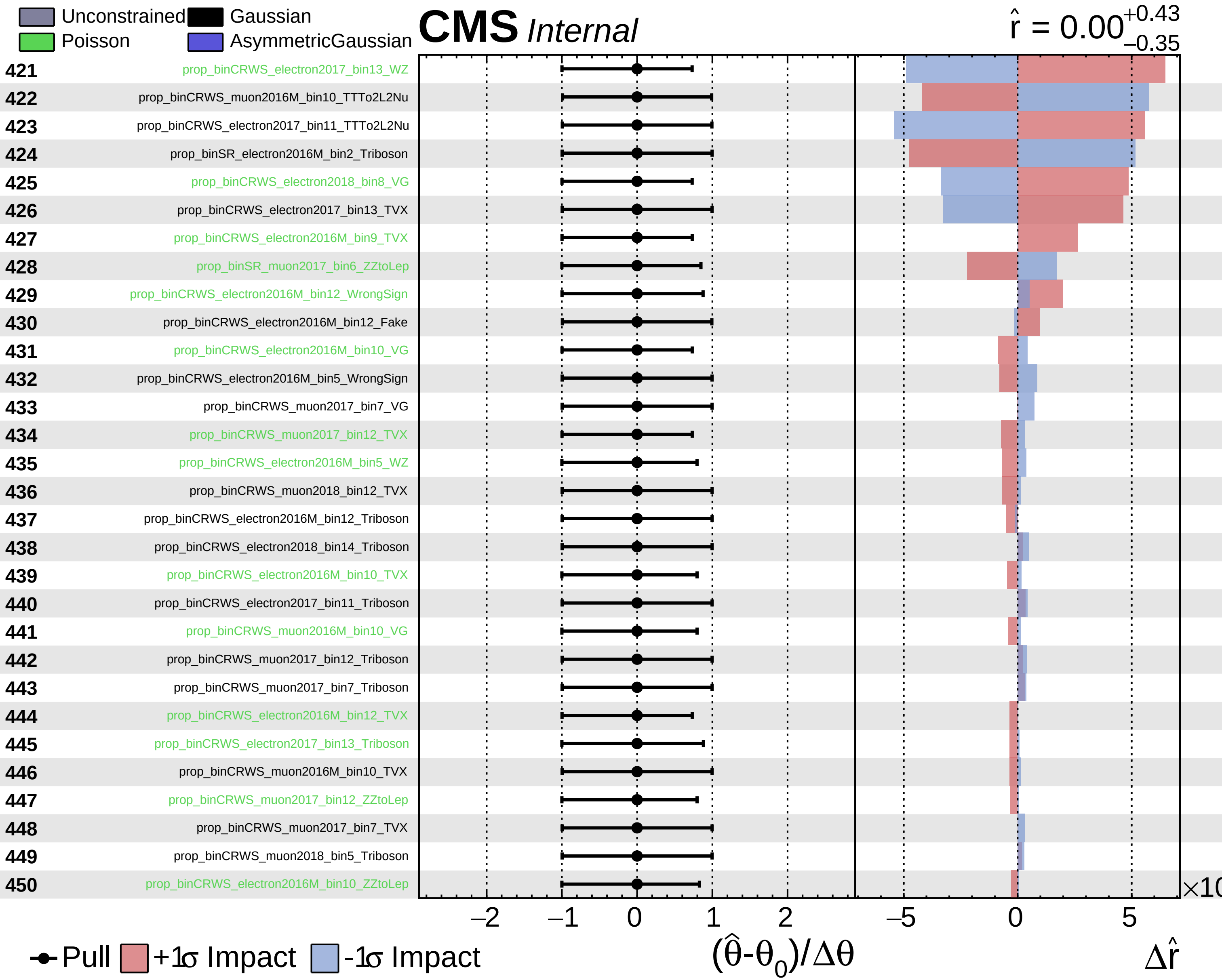
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.43}_{-0.35}$
331 prop_binCRTT_electron2016M_bin10
332 prop_binCRWS_electron2018_bin11
333 lep_2018
334 prop_binCRTT_muon2016M_bin5
335 prop_binCRWS_electron2016M_bin9_Fake
336 prop_binCRTT_electron2018_bin12
337 prop_binCRWS_muon2017_bin13
338 prop_binCRWS_muon2016M_bin14
339 prop_binCRTT_muon2017_bin9
340 prop_binCRTT_muon2016M_bin11
341 prop_binCRTT_electron2016M_bin15
342 prop_binCRTT_electron2016M_bin9
343 lep_2017
344 prop_binCRTT_electron2017_bin6
345 prop_binCRTT_muon2018_bin8
346 prop_binSR_muon2017_bin6_WrongSign
347 prop_binCRWS_electron2017_bin12
348 prop_binCRTT_muon2017_bin11
349 prop_binCRWS_muon2016M_bin10_Fake
350 prop_binCRWS_electron2018_bin14_Fake
351 prop_binCRTT_electron2016M_bin7
352 prop_binCRTT_electron2018_bin4
353 prop_binCRTT_muon2017_bin6
354 prop_binSR_electron2017_bin0
355 prop_binSR_muon2017_bin6_TTTo2L2Nu
356 prop_binCRTT_electron2018_bin10
357 prop_binCRWS_electron2017_bin13_Fake
358 prop_binCRWS_electron2016M_bin15
359 prop_binCRTT_muon2016M_bin1
360 prop_binCRTT_electron2016M_bin11
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



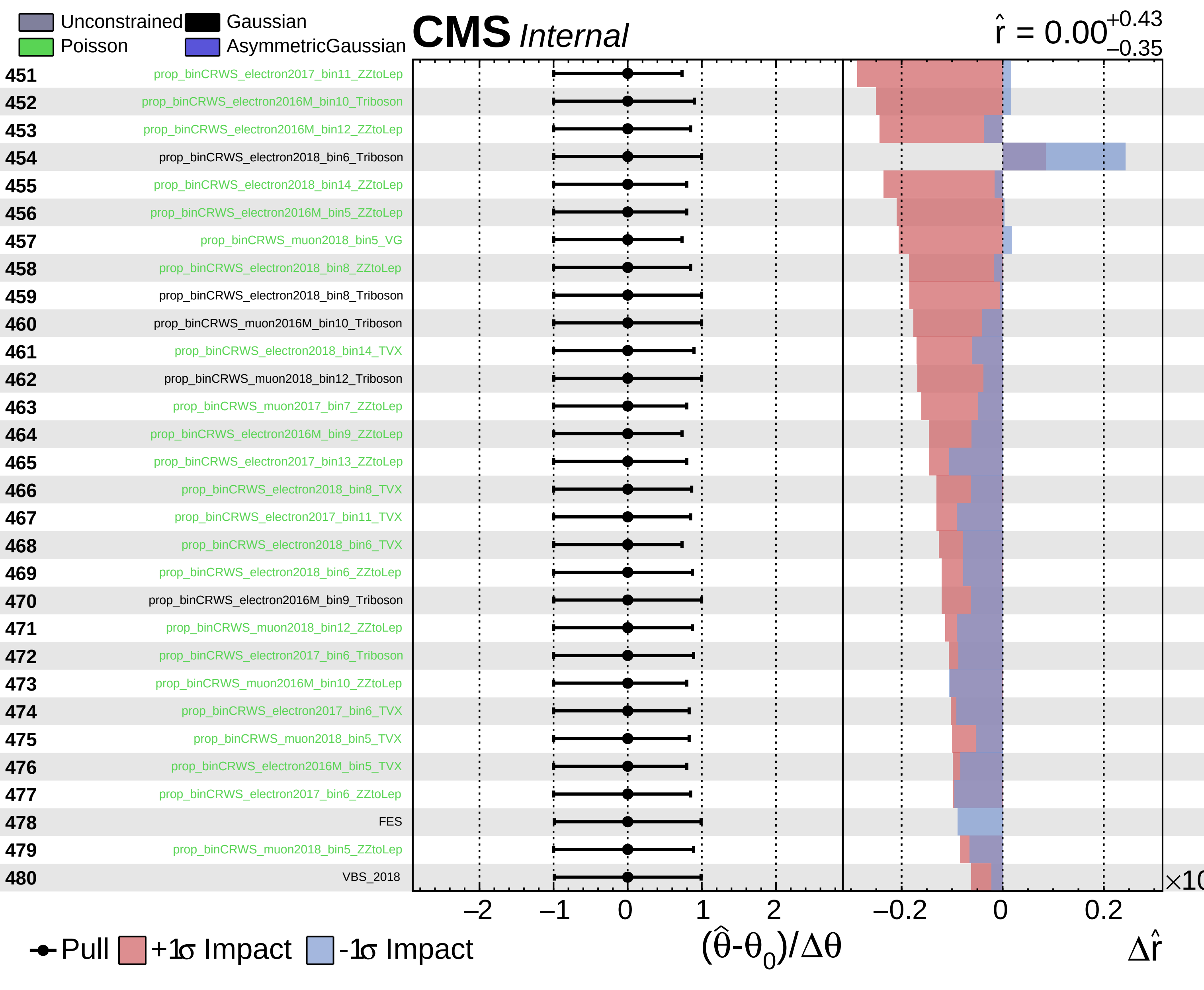
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.43}_{-0.35}$
361 prop_binSR_electron2016M_bin2_WZ
362 prop_binCRTT_muon2017_bin10
363 prop_binCRTT_muon2017_bin15
364 prop_binCRWS_electron2018_bin14_WrongSign
365 prop_binCRWS_muon2018_bin12_WrongSign
366 prop_binSR_electron2016M_bin3_VG
367 prop_binSR_electron2016M_bin2_WrongSign
368 prop_binSR_electron2016M_bin2_TTTo2L2Nu
369 prop_binSR_electron2016M_bin3_TTTo2L2Nu
370 prop_binCRTT_electron2016M_bin4
371 prop_binCRTT_muon2016M_bin13
372 prop_binCRWS_electron2017_bin14
373 prop_binCRTT_electron2018_bin6
374 prop_binCRWS_muon2018_bin12_TTTo2L2Nu
375 prop_binCRWS_electron2016M_bin10_WrongSign
376 prop_binCRTT_electron2017_bin8
377 prop_binCRWS_muon2017_bin15
378 prop_binCRWS_muon2017_bin12_Fake
379 prop_binCRWS_electron2018_bin14_TTTo2L2Nu
380 prop_binCRTT_electron2016M_bin5
381 prop_binCRWS_electron2016M_bin10_TTTo2L2Nu
382 lep_2016M
383 prop_binCRWS_muon2017_bin7_WrongSign
384 prop_binCRTT_muon2016M_bin7
385 prop_binCRTT_muon2016M_bin8
386 prop_binSR_electron2016M_bin3_ZZtoLep
387 prop_binCRWS_electron2016M_bin12_TTTo2L2Nu
388 prop_binCRWS_electron2016M_bin5_TTTo2L2Nu
389 prop_binCRWS_muon2017_bin12_TTTo2L2Nu
390 prop_binCRTT_muon2016M_bin14
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



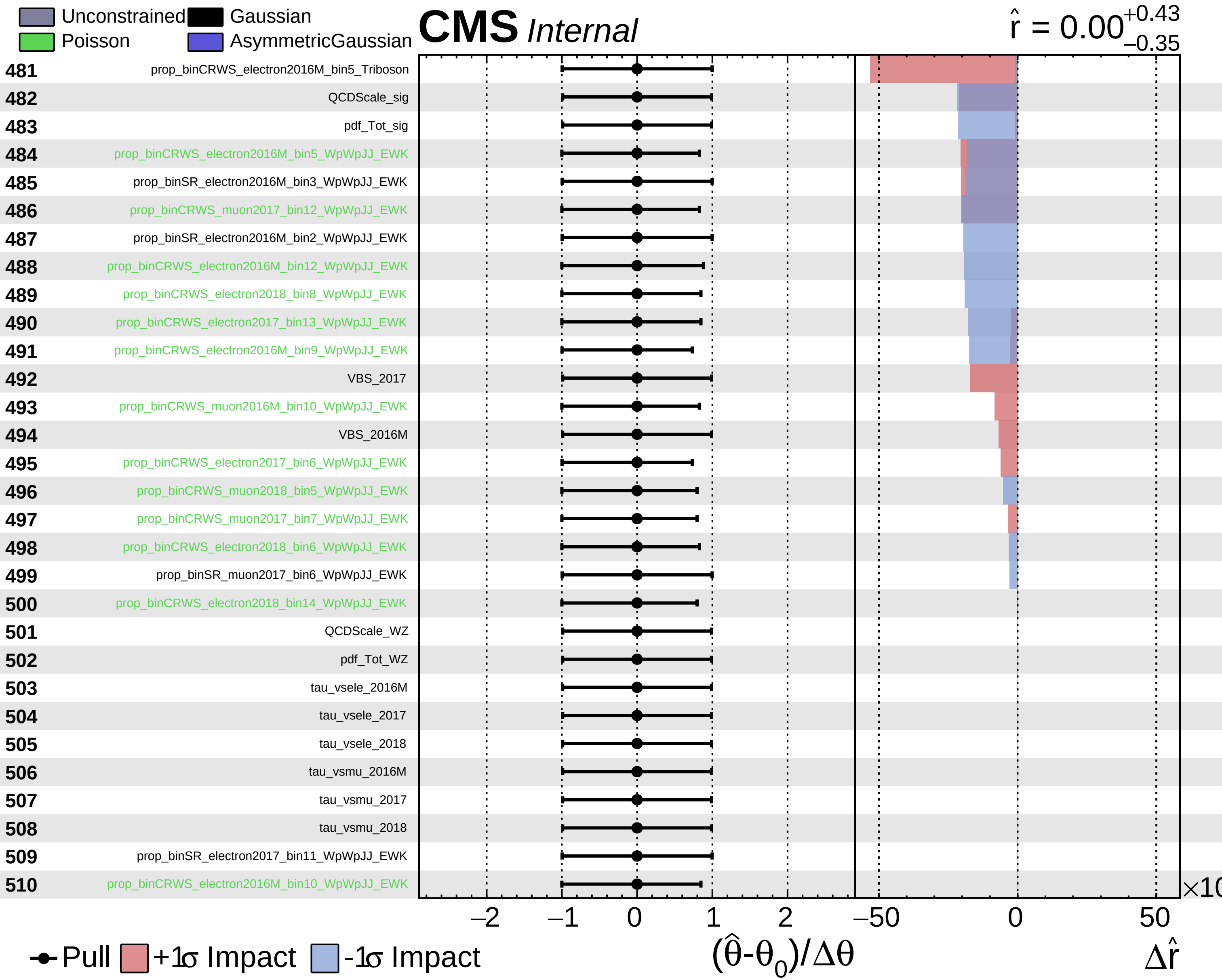
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.43}_{-0.35}$
391 prop_binSR_electron2016M_bin2_VG
392 prop_binCRWS_muon2017_bin12_WrongSign
393 prop_binSR_electron2016M_bin2_ZZtoLep
394 prop_binSR_electron2016M_bin3_WrongSign
395 prop_binCRTT_electron2017_bin14
396 prop_binSR_muon2017_bin6_WpWpJJ_QCD
397 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
398 prop_binSR_muon2017_bin6_TVX
399 prop_binSR_electron2016M_bin3_TVX
400 prop_binCRWS_muon2016M_bin10_WrongSign
401 prop_binCRWS_electron2017_bin13_WrongSign
402 prop_binCRWS_electron2018_bin6_WrongSign
403 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
404 prop_binCRWS_muon2018_bin5_WrongSign
405 prop_binSR_muon2017_bin6_Triboson
406 prop_binCRWS_electron2017_bin11_WrongSign
407 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
408 prop_binCRWS_electron2017_bin13_TTTo2L2Nu
409 prop_binCRWS_muon2017_bin7_TTTo2L2Nu
410 prop_binCRWS_electron2018_bin6_TTTo2L2Nu
411 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
412 prop_binCRWS_electron2018_bin8_WrongSign
413 prop_binCRWS_electron2017_bin6_WrongSign
414 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
415 prop_binCRWS_electron2018_bin8_TTTo2L2Nu
416 prop_binCRWS_electron2018_bin14_VG
417 prop_binCRWS_electron2017_bin6_WZ
418 prop_binSR_electron2016M_bin2_TVX
419 prop_binCRWS_electron2016M_bin9_WrongSign
420 prop_binSR_electron2016M_bin3_Triboson
−2 −1 0 1 2
−0.02 0 0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.43}_{-0.35}$
421 prop_binCRWS_electron2017_bin13_WZ
422 prop_binCRWS_muon2016M_bin10_TTTo2L2Nu
423 prop_binCRWS_electron2017_bin11_TTTo2L2Nu
424 prop_binSR_electron2016M_bin2_Triboson
425 prop_binCRWS_electron2018_bin8_VG
426 prop_binCRWS_electron2017_bin13_TVX
427 prop_binCRWS_electron2016M_bin9_TVX
428 prop_binSR_muon2017_bin6_ZZtoLep
429 prop_binCRWS_electron2016M_bin12_WrongSign
430 prop_binCRWS_electron2016M_bin12_Fake
431 prop_binCRWS_electron2016M_bin10_VG
432 prop_binCRWS_electron2016M_bin5_WrongSign
433 prop_binCRWS_muon2017_bin7_VG
434 prop_binCRWS_muon2017_bin12_TVX
435 prop_binCRWS_electron2016M_bin5_WZ
436 prop_binCRWS_muon2018_bin12_TVX
437 prop_binCRWS_electron2016M_bin12_Triboson
438 prop_binCRWS_electron2018_bin14_Triboson
439 prop_binCRWS_electron2016M_bin10_TVX
440 prop_binCRWS_electron2017_bin11_Triboson
441 prop_binCRWS_muon2016M_bin10_VG
442 prop_binCRWS_muon2017_bin12_Triboson
443 prop_binCRWS_muon2017_bin7_Triboson
444 prop_binCRWS_electron2016M_bin12_TVX
445 prop_binCRWS_electron2017_bin13_Triboson
446 prop_binCRWS_muon2016M_bin10_TVX
447 prop_binCRWS_muon2017_bin12_ZZtoLep
448 prop_binCRWS_muon2017_bin7_TVX
449 prop_binCRWS_muon2018_bin5_Triboson
450 prop_binCRWS_electron2016M_bin10_ZZtoLep
−2 −1 0 1 2
−5 0 5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.43}_{-0.35}$
451 prop_binCRWS_electron2017_bin11_ZZtoLep
452 prop_binCRWS_electron2016M_bin10_Triboson
453 prop_binCRWS_electron2016M_bin12_ZZtoLep
454 prop_binCRWS_electron2018_bin6_Triboson
455 prop_binCRWS_electron2018_bin14_ZZtoLep
456 prop_binCRWS_electron2016M_bin5_ZZtoLep
457 prop_binCRWS_muon2018_bin5_VG
458 prop_binCRWS_electron2018_bin8_ZZtoLep
459 prop_binCRWS_electron2018_bin8_Triboson
460 prop_binCRWS_muon2016M_bin10_Triboson
461 prop_binCRWS_electron2018_bin14_TVX
462 prop_binCRWS_muon2018_bin12_Triboson
463 prop_binCRWS_muon2017_bin7_ZZtoLep
464 prop_binCRWS_electron2016M_bin9_ZZtoLep
465 prop_binCRWS_electron2017_bin13_ZZtoLep
466 prop_binCRWS_electron2018_bin8_TVX
467 prop_binCRWS_electron2017_bin11_TVX
468 prop_binCRWS_electron2018_bin6_TVX
469 prop_binCRWS_electron2018_bin6_ZZtoLep
470 prop_binCRWS_electron2016M_bin9_Triboson
471 prop_binCRWS_muon2018_bin12_ZZtoLep
472 prop_binCRWS_electron2017_bin6_Triboson
473 prop_binCRWS_muon2016M_bin10_ZZtoLep
474 prop_binCRWS_electron2017_bin6_TVX
475 prop_binCRWS_muon2018_bin5_TVX
476 prop_binCRWS_electron2016M_bin5_TVX
477 prop_binCRWS_electron2017_bin6_ZZtoLep
478 FES
479 prop_binCRWS_muon2018_bin5_ZZtoLep
480 VBS_2018
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.43}_{-0.35}$
481 prop_binCRWS_electron2016M_bin5_Triboson
482 QCDScale_sig
483 pdf_Tot_sig
484 prop_binCRWS_electron2016M_bin5_WpWpJJ_EWK
485 prop_binSR_electron2016M_bin3_WpWpJJ_EWK
486 prop_binCRWS_muon2017_bin12_WpWpJJ_EWK
487 prop_binSR_electron2016M_bin2_WpWpJJ_EWK
488 prop_binCRWS_electron2016M_bin12_WpWpJJ_EWK
489 prop_binCRWS_electron2018_bin8_WpWpJJ_EWK
490 prop_binCRWS_electron2017_bin13_WpWpJJ_EWK
491 prop_binCRWS_electron2016M_bin9_WpWpJJ_EWK
492 VBS_2017
493 prop_binCRWS_muon2016M_bin10_WpWpJJ_EWK
494 VBS_2016M
495 prop_binCRWS_electron2017_bin6_WpWpJJ_EWK
496 prop_binCRWS_muon2018_bin5_WpWpJJ_EWK
497 prop_binCRWS_muon2017_bin7_WpWpJJ_EWK
498 prop_binCRWS_electron2018_bin6_WpWpJJ_EWK
499 prop_binSR_muon2017_bin6_WpWpJJ_EWK
500 prop_binCRWS_electron2018_bin14_WpWpJJ_EWK
501 QCDScale_WZ
502 pdf_Tot_WZ
503 tau_vsele_2016M
504 tau_vsele_2017
505 tau_vsele_2018
506 tau_vsmu_2016M
507 tau_vsmu_2017
508 tau_vsmu_2018
509 prop_binSR_electron2017_bin11_WpWpJJ_EWK
510 prop_binCRWS_electron2016M_bin10_WpWpJJ_EWK
−2
−1
0
1
2
−50
0
50
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$